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A small business invests \$9,900 in equipment to produce a product. Each unit of the product costs \$0.65 to produce and is sold for \$1.20. How many units of the product must be sold before the revenue received equals the total expense of production, including the initial investment in equipment?

A. 12,000

B. 14,500

C. 15,230

D. 18,000

E. 20,000

Arithmetic Rate

Let n be the number of units desired. Then, in dollars, the revenue received is $1.2n$ and the total expense is $9,900 + 0.65n$. These two amounts are equal when $1.2n = 9,900 + 0.65n$, or $0.55n = 9,900$. Therefore, $n = \frac{9,900}{0.55} = \frac{990,000}{55} = \frac{90,000}{5} = 18,000$.

The correct answer is D.

Estimated Number of Home-Schooled Students by State, January 2001

State	Number (in thousands)
A	181
B	125
C	103
D	79
E	72

•

According to the table shown, the estimated number of home-schooled students in State A is approximately what percent greater than the number in State D?

A. 25%

B. 55%

C. 100%

D. 125%

E. 155%

Arithmetic Percents

The percent increase from the number in State D to the number in State A is

$$= \left(\frac{181,000 - 79,000}{79,000} \times 100\right)\% \quad \text{expression for percent increase}$$

$$= \left(\frac{181 - 79}{79} \times 100\right)\% \quad \text{reduce fraction}$$

$$= \left(\frac{102}{79} \times 100\right)\% \quad \text{subtract}$$

$$\approx \left(\frac{100}{80} \times 100\right)\% \quad \text{approximate}$$

$$=(1.25 \times 100)\% \quad \text{divide}$$

$$= 125\% \quad \text{multiply}$$

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The correct answer is D.

•

When n liters of fuel were added to a tank that was already $\frac{1}{3}$ full, the tank was filled to $\frac{7}{9}$ of its capacity. In terms of n, what is the capacity of the tank, in liters?

A. $\frac{10}{9}n$

B. $\frac{4}{3}n$

C. $\frac{3}{2}n$

D. $\frac{9}{4}n$

E. $\frac{7}{3}n$

Algebra Applied Problems

Let C represent the capacity of the tank, in liters. It follows that

$$\begin{aligned} \frac{1}{3}C + n &= \frac{7}{9}C && \text{given} \\ n &= \frac{7}{9}C - \frac{1}{3}C && \text{subtract } \frac{1}{3}C \text{ from both sides} \\ n &= \frac{4}{9}C && \text{combine like terms} \\ \frac{9}{4}n &= C && \text{divide both sides by } \frac{4}{9} \end{aligned}$$

The correct answer is D.

•

A bicycle race on a mountain trail began at an elevation 850 meters above Town X. The race course consisted of 5 phases, with the following elevation changes: a descent of 280 meters, an ascent of 350 meters, a descent of 620 meters, an ascent of 100 meters, and a descent of 400 meters. At what elevation, in meters above or below Town X, did the race end?

A. 850 below Town X

B. Same elevation as Town X

C. 850 above Town X

D. 1,700 above Town X

E. 2,600 above Town X

Arithmetic Applied Problems

Starting at 850 meters above Town X and adjusting for the 5 elevation changes, we find that the race ended at an elevation of 850 − 280 + 350 − 620 + 100 − 400 = 0 meters above Town X; that is, at the same elevation as Town X.

The correct answer is B.

•

The harvest yield from a certain apple orchard was 350 bushels of apples. If x of the trees in the orchard each yielded 10 bushels of apples, what fraction of the harvest yield was from these x trees?

A $\frac{x}{}$

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A. $\overline{35}$

B. $1-\frac{x}{35}$

C. $10x$

D. $35-x$

E. $350-10x$

Algebra Algebraic Expressions

Since each of the x trees yielded 10 bushels, the total number of bushels yielded by these trees was $10x$. Since the yield of the entire orchard was 350 bushels, $\frac{10x}{350}=\frac{x}{35}$ was the fraction of the total yield from these x trees.

The correct answer is A.

•

In a certain fraction, the denominator is 16 greater than the numerator. If the fraction is equivalent to 80 percent, what is the denominator of the fraction?

A. 32

B. 64

C. 72

D. 80

E. 120

Algebra First-Degree Equations

Let n be the numerator of the fraction. Then $n+16$ is the denominator of the fraction. From the given information it follows that $\frac{n}{n+16}=80\%$, which is solved below.

$$\begin{aligned} \frac{n}{n+16} &= \frac{4}{5} && \text{convert 80\% to a fraction} \\ 5n &= 4(n+16) && \text{clear fractions} \\ 5n &= 4n+64 && \text{simplify} \\ n &= 64 && \text{subtract } 4n \text{ from both sides} \end{aligned}$$

Therefore, the denominator of the fraction is $n+16=64+16=80$.

The correct answer is D.

•

Jonathan, Kyle, and Michael agreed to pay equal shares of the monthly rent on an apartment. Because Jonathan has a cat, he paid a 1-time pet deposit of $250. The pet deposit and Jonathan's share of the monthly rent for the first month totaled $525. How much is the monthly rent on the apartment?

A. $1,575

B. $1,075

C. $1,025

D. $825

E. $775

Arithmetic Applied Problems

Jonathan's share of the monthly rent was $525 minus the $250 pet deposit, which comes to $275. Since his two roommates also agreed to pay equal $275 thirds of the monthly rent, the total monthly rent was 3 × $275 = $825.

The correct answer is D.

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•

The ratio of the amount of Alex's fuel oil bill for the month of February to the amount of his fuel oil bill for the month of January was $\frac{3}{2}$. If the fuel oil bill for February had been \$40 more, the corresponding ratio would have been $\frac{5}{3}$. How much was Alex's fuel oil bill for January?

A. \$240

B. \$300

C. \$360

D. \$450

E. \$540

Algebra Ratio and Proportion

Let x be the amount of Alex's fuel bill for January. Then the amount of his fuel bill for February is $\frac{3x}{2}$. We're also told that $\frac{3x}{2} + \$40 = \frac{5x}{3}$, so $\$40 = \frac{5x}{3} - \frac{3x}{2} = \frac{10x - 9x}{6} = \frac{x}{6}$. Thus, $x = 6(\$40) = \$240$.

The correct answer is A.

•

Al and Ben are drivers for SD Trucking Company. One snowy day, Ben left SD at 8:00 a.m. heading east and Al left SD at 11:00 a.m. heading west. At a particular time later that day, the dispatcher retrieved data from SD's vehicle tracking system. The data showed that, up to that time, Al had averaged 40 miles per hour and Ben had averaged 20 miles per hour. It also showed that Al and Ben had driven a combined total of 240 miles. At what time did the dispatcher retrieve data from the vehicle tracking system?

A. 1:00 p.m.

B. 2:00 p.m.

C. 3:00 p.m.

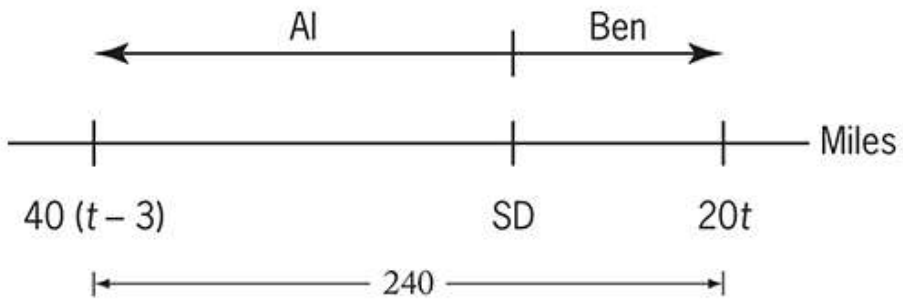
D. 5:00 p.m.

E. 6:00 p.m.

Algebra Applied Problems

Let t be the number of hours after 8:00 a.m. that Ben drove. Then Al, who began 3 hours after Ben began and thus drove 3 hours less than Ben, drove $(t-3)$ hours. First, find the total distance each drove in terms of t.

	rate	time	distance = rate × time
Al	40	$t-3$	$40(t-3)$
Ben	20	t	$20t$



Since the combined total distance that Al and Ben drove was 240 miles, it follows that $40(t-3) + 20t = 240$. Solve for t.

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$$\begin{aligned} 40(t-3)+20t &= 240 && \text{given} \\ 60t-120 &= 240 && \text{expand and combine like terms} \\ 60t &= 360 && \text{add 120 to both sides} \\ t &= 6 && \text{divide both sides by 60} \end{aligned}$$

Therefore, the data was retrieved 6 hours after 8:00 a.m., which was 2:00 p.m.

The correct answer is B.

•

If $f = 3r\left(\frac{s}{1{,}000}\right)^2$ and if $s = 2{,}000$, what is r in terms of f?

A. $\frac{f}{6}$

B. $\frac{f}{12}$

C. $\frac{f}{36}$

D. $\frac{f^2}{36}$

E. $\frac{3f}{4}$

Algebra Formulas

Since $f = 3r\left(\frac{2{,}000}{1{,}000}\right)^2 = 3r(2)^2 = 12r$, then $r = \frac{f}{12}$.

The correct answer is B.

•

At the start of an experiment, a certain population consisted of 3 animals. At the end of each month after the start of the experiment, the population size was double its size at the beginning of that month. Which of the following represents the population size at the end of 10 months?

A. 2^3

B. 3^2

C. $2(3^{10})$

D. $3(2^{10})$

E. $3(10^2)$

Arithmetic Applied Problems; Sequences

The population doubles each month, so multiply the previous month's population by 2 to get the next month's population. Thus, at the end of the 1st month the population will be (3)(2), at the end of the 2nd month the population will be (3)(2)(2), at the end of the 3rd month the population will be (3)(2)(2)(2), and so on. Therefore, at the end of the 10th month the population will be the product of 3 and ten factors of 2, which equals $3(2^{10})$.

The correct answer is D.

•

The formula $F = \frac{9}{5C} + 32$ gives the relationship between the temperature in degrees Fahrenheit, F, and the

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temperature given in degrees Celsius, C. If the temperature is 85 degrees Fahrenheit, what is the temperature, to the nearest degree, in degrees Celsius?

A. 18°C

B. 23°C

C. 29°C

D. 47°C

E. 51°C

Algebra Formulas

Using the given formula to convert 85 degrees Fahrenheit to Celsius, we get $85 = \frac{9}{5}\mathrm{C} + 32$, so $53 = \frac{9}{5}C$, which means $C = 53\left(\frac{5}{9}\right) = \frac{265}{9} = 29\frac{4}{9}$. So, rounding to the nearest degree Celsius, we find that 85 degrees Fahrenheit is 29 degrees Celsius.

The correct answer is C.

•

If x and y are positive integers such that y is a multiple of 5 and $3x + 4y = 200$, then x must be a multiple of which of the following?

A. 3

B. 6

C. 7

D. 8

E. 10

Arithmetic Properties of Numbers

Since y is a multiple of 5, consider the following table:

y	$4y$	$3x = 200 - 4y$	$x = \frac{200-4y}{3}$	Is x a positive integer?
5	20	180	60	**yes**
10	40	160	$\frac{160}{3}$	no
15	60	140	$\frac{140}{3}$	no
20	80	120	40	**yes**
25	100	100	$\frac{100}{3}$	no
30	120	80	$\frac{80}{3}$	no
35	140	60	20	**yes**
40	160	40	$\frac{40}{3}$	no
45	180	20	$\frac{20}{3}$	no
50	200	0	0	no

The only positive integers x that satisfy the $3x + 4y = 200$, where y is a multiple of 5, are 60, 40, and 20; each is a multiple of 10.

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multiple of 10.

Note: It is not necessary to perform all of the calculations shown here, nor is it necessary to fill out the table completely. Using "number sense" to see the patterns (e.g., numbers in the second column increase by 20 while numbers in the third column decrease by 20) will shorten the work considerably.

The correct answer is E.

•

Which of the following expressions can be written as an integer?

I. $\left(\sqrt{82}+\sqrt{82}\right)^2$

II. $(82)\left(\sqrt{82}\right)$

III. $\dfrac{\left(\sqrt{82}\right)\left(\sqrt{82}\right)}{82}$

A. None

B. I only

C. III only

D. I and II

E. I and III

Arithmetic Operations with Radical Expressions

Expression I represents an integer because $\left(\sqrt{82}+\sqrt{82}\right)^2=\left(2\sqrt{82}\right)^2=(4)(82)$. Expression II does not represent an integer because $(82)\ \sqrt{82}\ =\sqrt{82}^3$ and $82^3 = 2^3 \times 41^3$ is not a perfect square. Regarding this last assertion, note that the square of any integer has the property that each of its distinct prime factors is repeated an even number of times. For example, $24^2 = (2^3 \times 3)^2 = 2^6 \times 3^2$ has the prime factor 2 repeated 6 times and the prime factor 3 repeated twice. Expression III represents an integer, because $\dfrac{\left(\sqrt{82}\right)\left(\sqrt{82}\right)}{82}=\dfrac{82}{82}=1$.

The correct answer is E.

•

Which of the following is equal to the average (arithmetic mean) of $(x+2)^2$ and $(x-2)^2$?

A. x^2

B. x^2+2

C. x^2+4

D. x^2+2x

E. x^2+4x

Algebra Statistics

The average of $(x+2)^2$ and $(x-2)^2$ is

$$\frac{(x+2)^2+(x-2)^2}{2}=\frac{x^2+4x+4+x^2-4x+4}{2}=\frac{2x^2+8}{2}=x^2+4.$$

The correct answer is C.

•

If $x^2-2<0$, which of the following specifies all the possible values of x?

A. $0<x<2$

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B. $0 < x < \sqrt{2}$

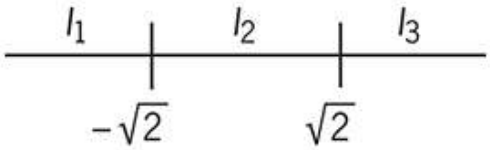
C. $-\sqrt{2} < x < \sqrt{2}$

D. $-2 < x < 0$

E. $-2 < x < 2$

Algebra Inequalities

The corresponding equality $x^2 - 2 = 0$ has two solutions, $x = \sqrt{2}$ and $x = -\sqrt{2}$, and thus there are three intervals to test for inclusion in the solution of the inequality: $x < -\sqrt{2}$, $-\sqrt{2} < x < \sqrt{2}$, and $x > \sqrt{2}$, labeled I_1, I_2, and I_3, respectively, in the figure below.



Next, choose a value from each of these intervals to test whether the inequality holds:

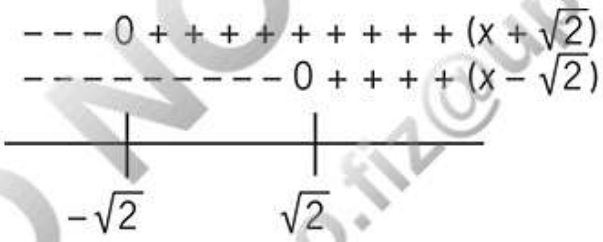
From I_1, for example, choose $x = -2$; the inequality becomes $(-2)^2 - 2 < 0$ (False).

From I_2, for example, choose $x = 0$; the inequality becomes $(0)^2 - 2 < 0$ (True).

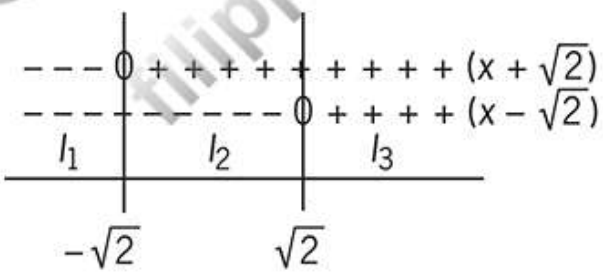
From I_3, for example, choose $x = 2$; the inequality becomes $(2)^2 - 2 < 0$ (False).

Therefore, the solution consists of only values in I_2.

Alternatively, the expression $x^2 - 2$ can be factored as $(x - \sqrt{2})(x + \sqrt{2})$. The value of the factor $(x - \sqrt{2})$ is 0 at $x = \sqrt{2}$, negative for values of x less than $\sqrt{2}$, and positive for values of x greater than $\sqrt{2}$. Similarly, the value of the factor $(x + \sqrt{2})$ is 0 at $x = -\sqrt{2}$, negative for values of x less than $-\sqrt{2}$, and positive for values of x greater than $-\sqrt{2}$. Using the number line, this information can be summarized as shown below:



Then, as shown below, vertical lines through the zeros partition the number line into three intervals I_1, I_2, and I_3.



I_1: $x < -\sqrt{2}$, where both factors are negative, so the product $(x - \sqrt{2})(x + \sqrt{2})$ is positive.

I_2: $-\sqrt{2} < x < \sqrt{2}$, where one factor is positive and the other is negative, so the product $(x - \sqrt{2})(x + \sqrt{2})$ is negative.

I_3: $x > \sqrt{2}$, where both factors are positive, so the product $(x - \sqrt{2})(x + \sqrt{2})$ is positive.

Therefore, the solution consists of only values in I_2.

The correct answer is C.

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Book Number	Pages in Book	Total Pages Read
1	253	253
2	110	363
3	117	480
4	170	650
5	155	805
6	50	855
7	205	1,060
8	70	1,130
9	165	1,295
10	105	1,400
11	143	1,543
12	207	1,750

•

Shawana made a schedule for reading books during 4 weeks (28 days) of her summer vacation. She has checked out 12 books from the library. The number of pages in each book and the order in which she plans to read the books are shown in the table above. She will read exactly 50 pages each day. The only exception will be that she will never begin the next book on the same day that she finishes the previous one, and therefore on some days she may read fewer than 50 pages. At the end of the 28th day, how many books will Shawana have finished?

A. 7

B. 8

C. 9

D. 10

E. 11

Arithmetic Operations with Integers

Book 1: 6 days—50 pages on each of Days 1–5, 3 pages on Day 6 [5(50) + 3 = 253]

Book 2: 3 days—50 pages on each of Days 7 and 8, 10 pages on Day 9 [2(50) + 10 = 110]

Book 3: 3 days—50 pages on each of Days 10 and 11, 17 pages on Day 12 [2(50) + 17 = 117]

Book 4: 4 days—50 pages on each of Days 13–15, 20 pages on Day 16 [3(50) + 20 = 170]

Book 5: 4 days—50 pages on each of Days 17–19, 5 pages on Day 20 [3(50) + 5 = 155]

Book 6: 1 day—50 pages on Day 21 [1(50) = 50]

Book 7: 5 days—50 pages on each of Days 22–25, 5 pages on Day 26 [4(50) + 5 = 205]

Book 8: 2 days—50 pages on Day 27, 20 pages on Day 28 [50 + 20 = 70]

At this point, Shawana has read on a total of 28 days and has finished 8 books.

The correct answer is B.

•

In Western Europe, x bicycles were sold in each of the years 1990 and 1993. The bicycle producers of Western Europe had a 42 percent share of this market in 1990 and a 33 percent share in 1993. Which of the following represents the decrease in the annual number of bicycles produced and sold in Western Europe from 1990 to 1993?

A. 9% of $\frac{x}{100}$

B. 14% of $\frac{x}{100}$

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100

C. 75% of $\frac{x}{100}$

D. 9% of x

E. 14% of x

Arithmetic Percents

Of the x bicycles sold in Western Europe in 1990, 42% of them were produced in Western Europe. It follows that the number of bicycles produced and sold in Western Europe in 1990 was $0.42x$. Similarly, of the x bicycles sold in Western Europe in 1993, 33% were produced in Western Europe. It follows that the number of bicycles produced and sold in Western Europe in 1993 was $0.33x$. Therefore, the decrease in the annual number of bicycles produced and sold in Western Europe from 1990 to 1993 was $0.42x - 0.33x = 0.09x$, which is 9% of x.

The correct answer is D.

•

If k is a positive integer, what is the remainder when $(k+2)(k^3-k)$ is divided by 6?

A. 0

B. 1

C. 2

D. 3

E. 4

Algebra Properties of Numbers

Since k can be any positive integer, the remainder must be the same regardless of the value of k. If $k = 2$, for example, then $(k+2)(k^3-k) = (2+2)(2^3-2) = (4)(6)$, which is a multiple of 6, and therefore, the remainder when divided by 6 is 0.

Alternatively, factor the given expression:

$$\begin{aligned}(k+2)(k^3-k) &= (k+2)(k)(k^2-1)\\ &= (k+2)(k)(k+1)(k-1)\end{aligned}$$

Now, rearrange the factors in ascending order $(k-1)(k)(k+1)(k+2)$, and observe that for any positive integer k, the factors are four consecutive integers, two of which are even and one of which is divisible by 3. Therefore, $(k+2)(k^3-k)$ is divisible by both 2 and 3. Thus, $(k+2)(k^3-k)$ is divisible by 6 with 0 remainder.

The correct answer is A.

•

Which of the following fractions is closest to $\frac{1}{2}$?

A. $\frac{4}{7}$

B. $\frac{5}{9}$

C. $\frac{6}{11}$

D. $\frac{7}{13}$

E. $\frac{9}{16}$

Arithmetic Fractions

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Arithmetic Fractions

A fraction equals $\frac{1}{2}$ if its numerator is half its denominator. Thus, to compare the answer choices with $\frac{1}{2}$, rewrite $\frac{1}{2}$ to have the same denominators as the answer choices. This will allow for an easy determination of the distance to $\frac{1}{2}$.

answer choice	value	rewrite of $\frac{1}{2}$	distance to $\frac{1}{2}$
A	$\frac{4}{7}$	$\frac{3.5}{7}$	$\frac{0.5}{7}$
B	$\frac{5}{9}$	$\frac{4.5}{9}$	$\frac{0.5}{9}$
C	$\frac{6}{11}$	$\frac{5.5}{11}$	$\frac{0.5}{11}$
D	$\frac{7}{13}$	$\frac{6.5}{13}$	$\frac{0.5}{13}$
E	$\frac{9}{16}$	$\frac{8}{16}$	$\frac{1}{16}$

Clearly, $\frac{0.5}{13}$ is smaller than each of $\frac{0.5}{7}$, $\frac{0.5}{9}$, and $\frac{0.5}{11}$ because the numerators are equal, and thus larger denominators correspond to smaller values. Also, $\frac{0.5}{13} = \frac{1}{26}$ is smaller than $\frac{1}{16}$. Therefore, $\frac{7}{13}$ is closest to $\frac{1}{2}$.

The correct answer is D.

•

If $p \neq 0$ and $p - \frac{1-p^2}{p} = \frac{r}{p}$, then $r =$

A. $p + 1$

B. $2p - 1$

C. $p^2 + 1$

D. $2p^2 - 1$

E. $p^2 + p - 1$

Algebra Simplifying Algebraic Expressions

$$p - \frac{1-p^2}{p} = \frac{r}{p} \quad \text{given}$$
$$p^2 - (1-p^2) = r \quad \text{multiply both sides by } p$$
$$2p^2 - 1 = r \quad \text{combine like terms}$$

The correct answer is D.

•

If $\frac{|z|}{w} = 1$, which of the following must be true?

A. $z = -w$

B. $z = w$

C. $z^2 = w^2$

D. $z^3 = w^3$

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D. $z^2 = w^3$

E. $z^3 = w^3$

Algebra Absolute Value

If $\frac{|z|}{w} = 1$, then $|z| = w$. That is, either $z = w$ or $-z = w$. In either case, $z^2 = w^2$, so option C is correct. Option A is wrong because if z and w are both positive, then $z \neq -w$. Option B is wrong because if z is negative and w is positive, then $z \neq w$. To see that options D and E are wrong, suppose $z = -2$, so that $w = 2$. In that case $z^2 = 4 \neq w^3 = 8$, contrary to option D, and $z^3 = -8 \neq w^3 = 8$, contrary to option E.

The correct answer is C.

•

What number is 108 more than two-thirds of itself?

A. 72

B. 144

C. 162

D. 216

E. 324

Algebra First-Degree Equations

Let x be the number that is 108 more than two-thirds of itself. Then, $108 + \frac{2}{3}x = x$. Solve for x as follows:

$$108 + \frac{2}{3}x = x$$
$$108 = \frac{1}{3}x$$
$$324 = x$$

The correct answer is E.

•

A service provider charges c dollars for the first 50 hours of service used per month and 40 cents for each 30 minutes in excess of 50 hours used during the month. If x is an integer greater than 50, which of the following expressions gives this service provider's charge, in dollars, for a month in which x hours of service were used?

A. $c + 0.40x$

B. $c + 0.80x$

C. $c + 0.40(x - 50)$

D. $c + 0.80(x - 50)$

E. $c + 0.40(2x - 50)$

Algebra Algebraic Expressions

For x hours of service used in the month, the charge is c dollars for the first 50 hours used, leaving $(x - 50)$ hours to be charged at 40 cents per half hour, which is equivalent to 80 cents per hour. Therefore, the total charge, in dollars, for a month in which x hours of service were used is $c + 0.80(x - 50)$.

The correct answer is D.

•

A salesperson who had been driving at a speed of 100 kilometers per hour slowed down to a speed of 47 kilometers per hour. Approximately how many miles per hour was the speed reduced? (1 kilometer ≈ 0.625 mile)

A. 29

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A. 29

B. 33

C. 53

D. 63

E. 75

Arithmetic Measurement Conversion

$$(100-47)\frac{\text{km}}{\text{hr}} \quad \text{amount of reduction}$$

$$53\frac{\text{km}}{\text{hr}} \quad \text{subtract}$$

$$\left(53\frac{\cancel{\text{km}}}{\text{hr}}\right)\left(0.625\frac{\text{mi}}{\cancel{\text{km}}}\right) \quad 1 \text{ km} = 0.625 \text{ mi}$$

$$(50)(0.6)\frac{\text{mi}}{\text{hr}} \quad \text{estimate}$$

$$30\ \frac{\text{mi}}{\text{hr}} \quad \text{multiply}$$

From the calculations it follows that speed was reduced, in miles per hour, by a little more than 30; among the answer choices, only 33 is a reasonable choice. The exact value is 33.125, and this can be found by multiplying 53 and 0.625 or, since $(53)\left(\frac{625}{1{,}000}\right) = (53)\left(\frac{5}{8}\right) = \frac{265}{8}$, dividing 265 by 8.

The correct answer is B.

•

If $5x-8=12$ and $x=y+3$, then $y=$

A. $\frac{4}{5}$

B. 1

C. $\frac{17}{5}$

D. 4

E. 7

Algebra Simultaneous Equations

If $5x-8=12$, then $5x=20$, so $x=4$. Thus, if $x=y+3$, then $y=4-3=1$.

The correct answer is B.

	Recording Time	Viewing Time
Tuesday	4 hours	None
Wednesday	None	1 to 2 hours
Thursday	2 hours	None
Friday	None	2 to 3 hours

•

The table above shows the numbers of hours of television programs that Jane recorded last week and the numbers of hours she spent viewing these recorded programs. No recorded program was viewed more than once. If h is the number of hours of recorded programs she had not yet viewed by the end of Friday, which of the following intervals represents all of the possible values of h?

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represents all of the possible values of h?

A. $0 \leq h \leq 1$

B. $1 \leq h \leq 2$

C. $2 \leq h \leq 3$

D. $0 \leq h \leq 2$

E. $1 \leq h \leq 3$

Arithmetic Inequalities

By the end of Friday, Jane had recorded a total of 4 + 2 = 6 hours of programs, and she had viewed between 1 + 2 = 3 hours and 2 + 3 = 5 hours of these programs. Therefore, the number of hours of recorded programs that Jane had not yet viewed by the end of Friday was between 6 − 5 = 1 and 6 − 3 = 3.

The correct answer is E.

•

A dance troupe has a total of 50 dancers split into 2 groups. The costumes worn by Group A cost $80 each, and those worn by Group B cost $90 each. If the total cost of all the costumes is $4,270, what is the total cost of the costumes worn by Group B?

A. $1,840

B. $2,070

C. $2,135

D. $2,160

E. $2,430

Algebra Applied Problems; First-Degree Equations

Let n be the number of dancers in Group B. Then the number of dancers in Group A is $50 - n$, and we are given that $(50 - n)(\$80) + (n)(\$90) = \$4,270$.

$$\begin{aligned} (50-n)(80) + 90n &= 4,270 && \text{given} \\ (50-n)(8) + 9n &= 427 && \text{divide both sides by 10} \\ 400 - 8n + 9n &= 427 && \text{distributive law} \\ n &= 27 && \text{combine like terms and subtract 400 from each side} \end{aligned}$$

Therefore, the total cost of the costumes worn by Group B is 27($90) = $2,430.

The correct answer is E.

•

A doctor prescribed 18 cubic centimeters of a certain drug to a patient whose body weight was 120 pounds. If the typical dosage is 2 cubic centimeters per 15 pounds of body weight, by what percent was the prescribed dosage greater than the typical dosage?

A. 8%

B. 9%

C. 11%

D. 12.5%

E. 14.8%

Arithmetic Percents

If the typical dosage is 2 cubic centimeters per 15 pounds of body weight, then the typical dosage for a person who weighs 120 pounds is $2\left(\frac{120}{15}\right) = 2(8) = 16$ cubic centimeters. The prescribed dosage of 18 cubic centimeters is, therefore, $\left(\left(\frac{18-16}{\ }\right) \times 100\right)$% or 12.5% greater than the typical dosage.

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therefore, $\left(\left(\frac{\quad}{16}\right) \times 100\right)$% or 12.5% greater than the typical dosage.

The correct answer is D.

•

The function f is defined by $f(x) = \sqrt{x} - 10$ for all positive numbers x. If $u = f(t)$ for some positive numbers t and u, what is t in terms of u?

A. $\sqrt{\sqrt{u} + 10}$

B. $(\sqrt{u} + 10)^2$

C. $\sqrt{u^2 + 10}$

D. $(u + 10)^2$

E. $(u^2 + 10)^2$

Algebra Functions

Because $f(x) = \sqrt{x} - 10$, it follows that $f(t) = \sqrt{t} - 10$. The problem states that $u = f(t)$, so by substitution, $u = \sqrt{t} - 10$. To express t in terms of u, isolate t on one side of the equation:

$$\begin{aligned} u &= \sqrt{t} - 10 && \text{given} \\ u + 10 &= \sqrt{t} && \text{add 10 to both sides} \\ (u + 10)^2 &= t && \text{square both sides} \end{aligned}$$

Therefore, t in terms of u is $(u + 10)^2$.

The correct answer is D.

•

If 35% of a number is decreased by 15, then the result is 25% of the original number. What is the original number?

A. 1.5

B. 9

C. 25

D. 150

E. 900

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